

Hammerstein Integral Equation

$\Omega \subseteq \mathbb{R}^n$ measurable

$K: \Omega \times \Omega \rightarrow \mathbb{R}$ kernel function

$f: \Omega \times \mathbb{R} \rightarrow \mathbb{R}$ Carathéodory

$$x(s) + \int_{\Omega} k(s, t) f(t, x(t)) dt = y(s)$$

$[Kx](s) = \int_{\Omega} k(s, t) x(t) dt$ (linear operator)

$[N_f x](s) = f(s, x(s))$ (Nemytskii operator)